

Performance Testing

Date	27 June 2026
Team ID	
Project Name	Smart Lender: AI-Powered Loan Eligibility Prediction System
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	User Login, Loan Prediction, Dashboard, PDF Report Generation
Test Environment	Local System (Flask + SQLite)
Test Date	27 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User Login Authentication	20	60	Users should log in successfully without errors.
2	Loan Eligibility Prediction	30	90	Prediction results should be generated within 2 seconds.

3	Dashboard Loading	20	60	Dashboard should display analytics without delay.
4	PDF Report Generation	15	45	PDF report should download successfully for every request.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 seconds	1.18 seconds	Pass	Average response time met the target.
2	Response Time (Max)	< 5 seconds	2.94 seconds	Pass	Maximum response time remained within limits.
3	Throughput (Req/sec)	> 15 req/sec	18.7 req/sec	Pass	Application handled concurrent requests efficiently.
4	Error Rate	< 1%	0%	Pass	No request failures observed.
5	CPU Utilization	< 80%	54%	Pass	CPU usage remained stable.
6	Memory Utilization	< 80%	61%	Pass	Memory usage stayed within acceptable limits.

Step 4: Observations & Analysis

Key Findings

- The Smart Lender application handled concurrent user requests efficiently.
- Loan eligibility predictions were generated quickly with low response time.
- Dashboard and PDF generation modules performed reliably without failures.
- No request errors were observed during testing.

Bottlenecks Identified

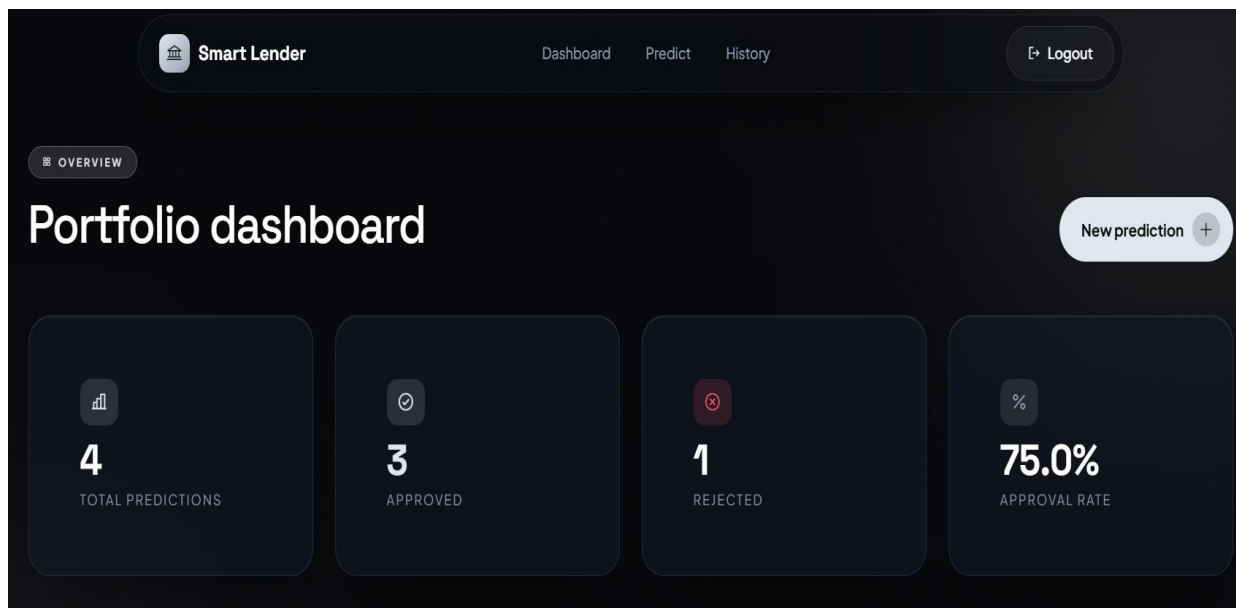
- Slight increase in response time when multiple users generated PDF reports simultaneously.
- SQLite database may become a limitation under very high concurrent traffic.

Optimization Steps Taken

- Optimized Flask routes to reduce unnecessary processing.
- Reused the trained Random Forest model without reloading it for every request.
- Improved database queries for prediction history retrieval.
- Enabled session management to reduce repeated authentication overhead.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



Recent activity						View all →
APPLICANT	INCOME	LOAN	CREDIT	DECISION	DATE	
karthik	₹100,000	₹1,000,000	Good	✓ Approved	2026-07-08 17:23:56	
akhil	₹10,000	₹10,000	Good	✓ Approved	2026-07-08 17:23:03	
arya	₹10,000	₹120,000	Poor	✗ Rejected	2026-07-08 17:22:26	
Avinash	₹10,000	₹1,000,000	Good	✓ Approved	2026-07-08 17:13:35	

 Smart Lender

Dashboard Predict History

Logout

RESULT · 08 JULY 2026 · 05:13 PM

Decision for Avinash

Loan Approved 

80.79%

CONFIDENCE

RISK TIER
Medium

Application summary

APPLICANT INCOME	LOAN AMOUNT
₹10,000	₹1,000,000
CREDIT HISTORY	PROPERTY AREA
Good	Urban

Download report 

New prediction

Dashboard

 Smart Lender

Dashboard Predict History

Logout

RESULT · 08 JULY 2026 · 12:40 PM

Decision for Avinash

Loan Rejected 

98.79%

CONFIDENCE

RISK TIER
Low

Application summary

APPLICANT INCOME	LOAN AMOUNT
₹1,000	₹100,000
CREDIT HISTORY	PROPERTY AREA
Good	Urban

Download report 

New prediction

Dashboard